

Michael Cusack-Nelkin

mc5882@columbia.edu | 646-957-3345 | michaelcusacknelkin.github.io | U.S. Citizen

Education

Columbia University | M.A. in Mathematics of Finance | New York, NY | Aug 2025 - Dec 2026

- GPA: 3.57 / 4.00
- Coursework: Statistical Aspects of Finance, Linear Regression Models, Neural Networks and Deep Learning

The College of William & Mary | B.S. in Physics | Williamsburg, VA | Aug 2018 - May 2022

- GPA: 3.72 / 4.00
- Minors in Data Science and Mathematics
- Coursework: Advanced Applied Machine Learning, Quantum Mechanics I & II, Fluid Mechanics
- Honors: Dean's List, *magna cum laude*

Work Experience

Deloitte Consulting | Business Technology Analyst - State of Texas | Austin, TX | Oct 2022 - Jun 2025

- Led a team of four software developers and testers shipping a fraud recovery system for the Texas Office of the Inspector General. Our work led to the client recovering \$10,000,000 of previously unrecoverable fraudulent claims.
- Presented technical work to clients on a daily basis, maintaining buy-in throughout the development and testing process.
- Owned the drafting and maintenance of a source-of-truth distributed systems plan, as well as hardware distribution processes, as the project grew from 30 to 400+ team members and the scope of work shifted.

DLA Piper | Financial Analyst Intern | New York, NY | Jun 2020 - Sept 2020

- Maintained tracking of profit and billing data for DLA Piper's finance practice.
- Provided weekly reports to leadership, including detailed profitability trends extracted from the firm's internal financial models.

Research and Publications

High Sharpe, Low Capacity: Optimally Trading the Nasdaq Closing Auction Imbalance | Summer 2026

- Built and backtested a statistical arbitrage system around the Nasdaq closing auction imbalance based on single-period optimization with an alpha forecast, covariance model, and instantaneous impact model. The forecast holds a 29.6% out-of-sample rank IC through rolling cross-validation, and the strategy backtests to a Sharpe of 8.53 at \$91.3M GMV. The strategy is subject to tight capacity and latency limits that make it best suited to an execution pipeline serving a larger portfolio.

Adaptive Execution Strategies Lead to Robust Improvements Over Naive Benchmark | Spring 2026

- Designed an adaptive execution strategy using L2 orderbook data and an execution simulator to test it. The simulator tracked the costs and price impact of any generic fill strategy comprising market orders, resting limit orders, and IOC-limit orders under variable latency and book resilience (modeled as a Poisson process). The final execution strategy used orderbook imbalance to adapt meta-order fill aggression and offered a large improvement over naive approaches.

Competitions

Millennium x Columbia University Trading Competition | New York, NY

- Led a team of four to win a Millennium Management ETF arbitrage competition for Columbia students.
- The team achieved low PnL with a modest simulated Sharpe ratio of 1.5 to win the competition.